

Project-III: Standard Cell Design Report

Team: Bit&Byte - AM

Course: EE619

April 18, 2026

1 Team and Work Distribution

This project was executed by a two-member team. The design, layout, and characterization of the standard cells were divided as follows to ensure clear roles and responsibilities:

- **Mahamkali Murari:** Responsible for the complete design, layout, and characterization of the **NAND4** standard cell.
- **Arava Prema Chandra Reddy:** Responsible for the complete design, layout, and characterization of the **NOR4** standard cell.

2 Reference CMOS Inverter

Before designing the complex standard cells, a reference CMOS inverter was designed to establish the baseline worst-case propagation delay. To achieve highly symmetric rise and fall times in the gpdk 180nm process, the PMOS transistor was sized appropriately to compensate for the lower mobility of holes compared to electrons.

The chosen transistor sizes and corresponding delays for the reference inverter are:

- **Length (L):** 360 nm
- **NMOS Width (W_n):** 720 nm
- **PMOS Width (W_p):** 2160 nm
- **Propagation Delay t_{pLH} :** 0.036 ns
- **Propagation Delay t_{pHL} :** 0.038 ns

This sizing logic ensures that the worst-case equivalent resistance of our standard cells can be scaled to match this highly symmetric baseline inverter.

3 First Standard Cell: NAND4

Designer: Mahamkali Murari

3.1 Schematic and Transistor Sizing

The NAND4 gate ($F = \overline{A \cdot B \cdot C \cdot D}$) was implemented in a single stage using complementary static CMOS logic.

To match the worst-case delay of the reference inverter, the transistor sizes were scaled as follows:

- **NMOS Sizing:** The pull-down network consists of 4 NMOS transistors in series. To maintain the same equivalent resistance as the reference inverter's single NMOS, the width was scaled by a factor of 4. Therefore, NMOS Width = 2880 **nm**.
- **PMOS Sizing:** The pull-up network consists of 4 PMOS transistors in parallel. In the worst-case scenario, only one PMOS turns on. Therefore, the PMOS width remains the same as the reference inverter. PMOS Width = 2160 **nm**.

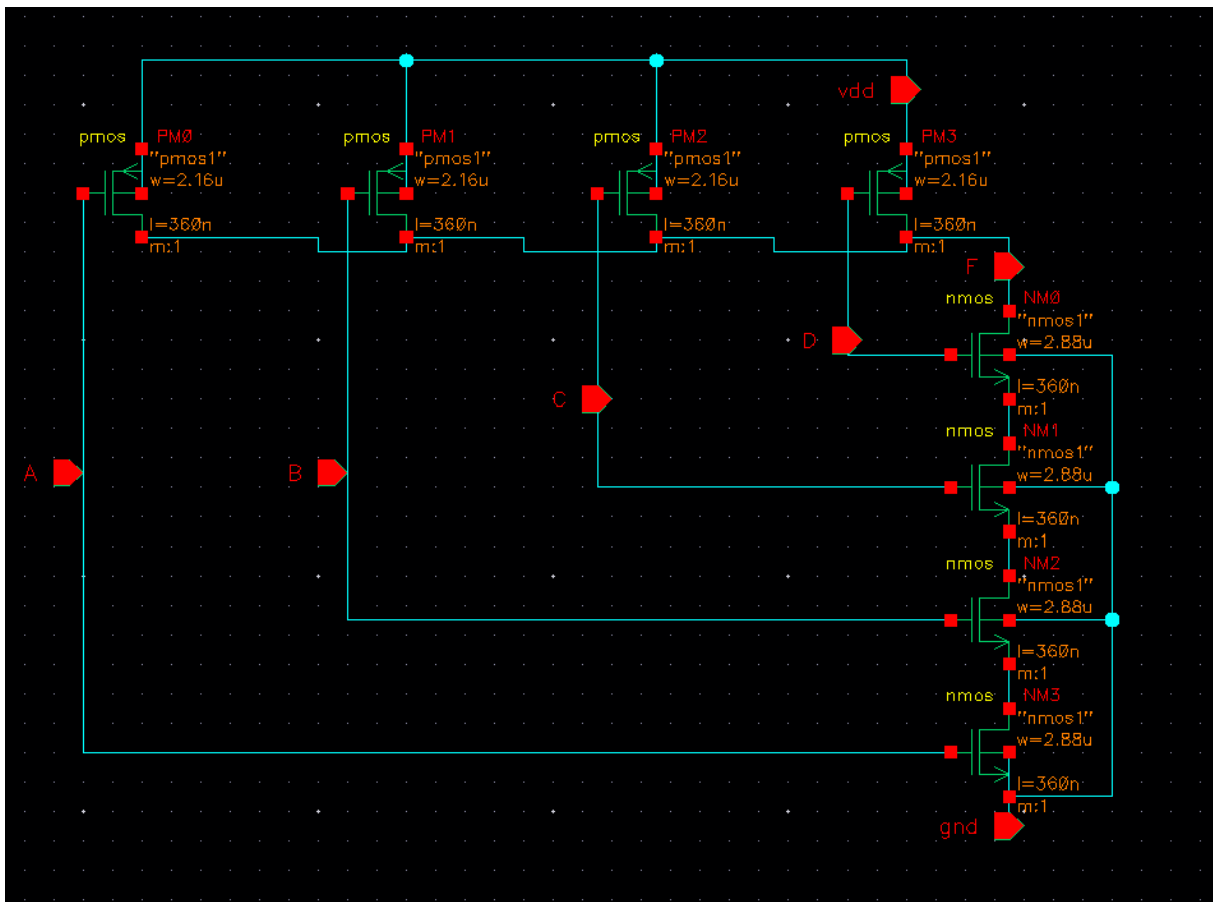


Figure 1: Schematic of the NAND4 standard cell.

3.2 Layout and Verification

The layout was designed adhering to the team’s established cell height and VDD/GND rail widths.

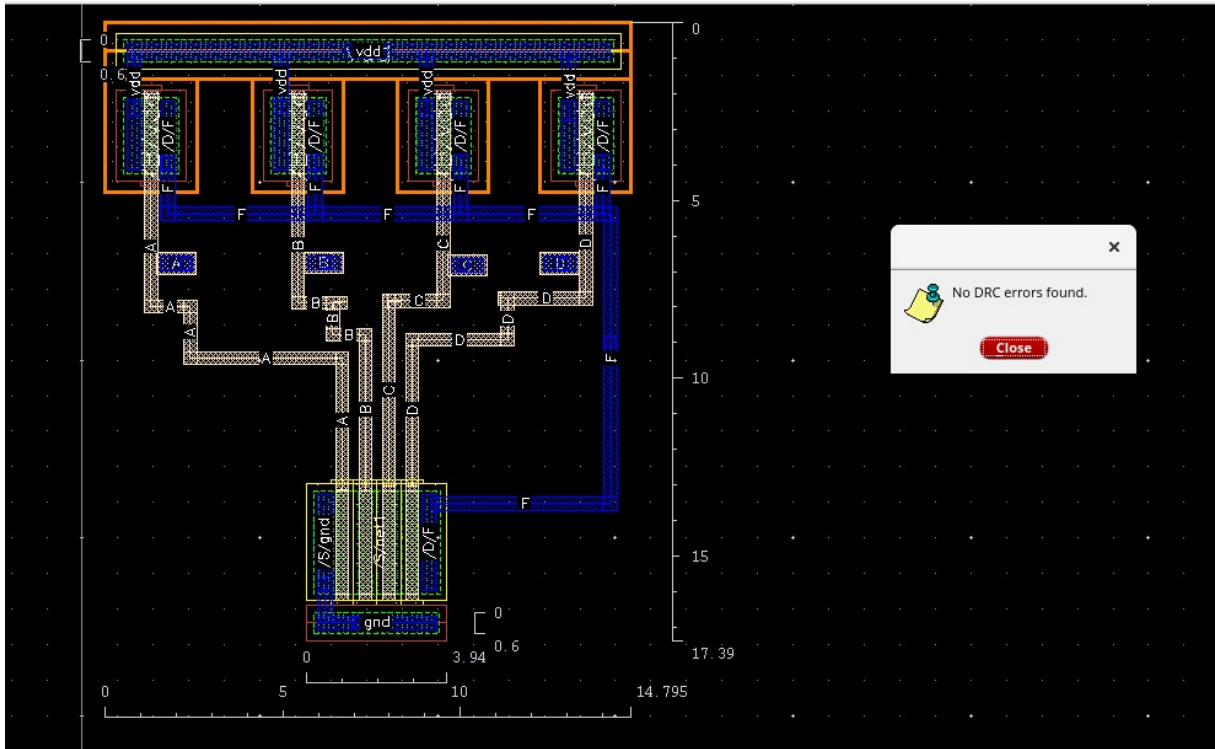


Figure 2: Layout of the NAND4 cell with clean DRC log. Height and width are explicitly marked.

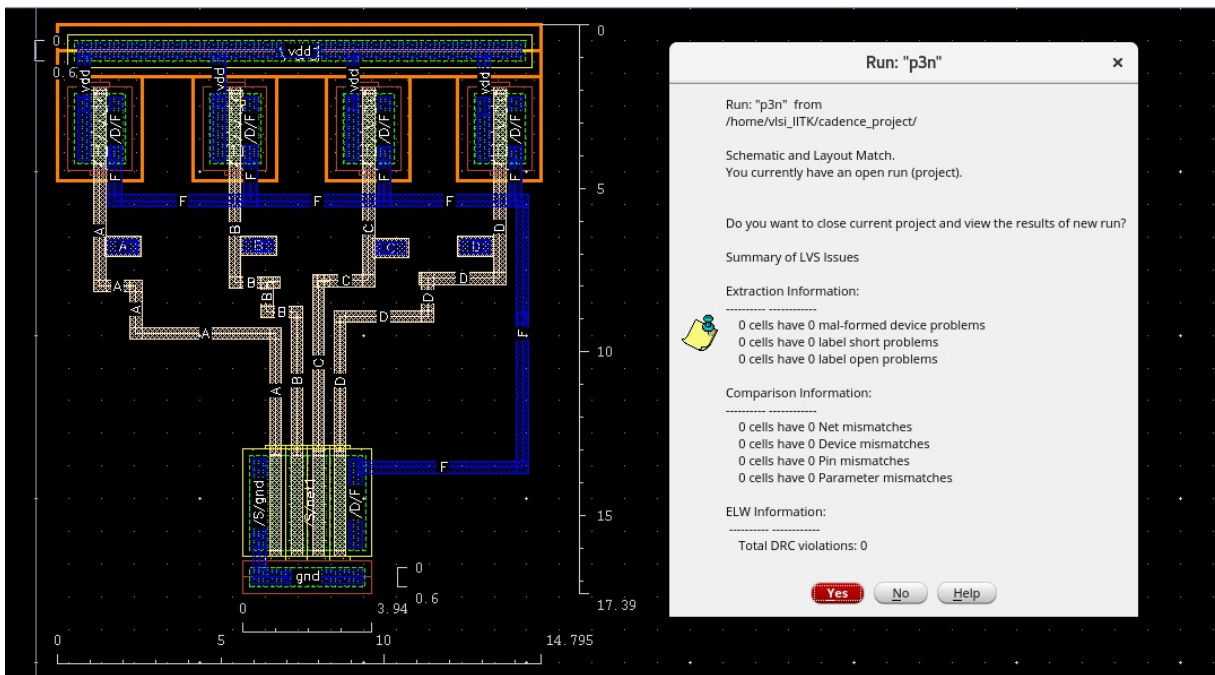


Figure 3: Layout of the NAND4 cell with clean LVS (Layout vs Schematic) log.

3.3 Propagation Delay Analysis

3.3.1 Schematic Simulation Delays

t_{pLH} from A to F ($ABCD : 1111 \rightarrow 0111$)			t_{pHL} from A to F ($ABCD : 0111 \rightarrow 1111$)		
Load capacitance	10 fF	50 fF	Load capacitance	10 fF	50 fF
Input transition			Input transition		
20 ps	0.326 ns	0.505 ns	20 ps	0.143 ns	0.245 ns
100 ps	0.336 ns	0.514 ns	100 ps	0.153 ns	0.256 ns

Table 1: Propagation delay from A to F in NAND4 based on schematic simulations.

3.3.2 Post-Layout (PEX) Simulation Delays

t_{pLH} from A to F ($ABCD : 1111 \rightarrow 0111$)			t_{pHL} from A to F ($ABCD : 0111 \rightarrow 1111$)		
Load capacitance	10 fF	50 fF	Load capacitance	10 fF	50 fF
Input transition			Input transition		
20 ps	0.310 ns	0.485 ns	20 ps	0.145 ns	0.247 ns
100 ps	0.321 ns	0.496 ns	100 ps	0.154 ns	0.256 ns

Table 2: Propagation delay from A to F in NAND4 based on post-layout simulations.

3.4 Transient Analysis Waveforms

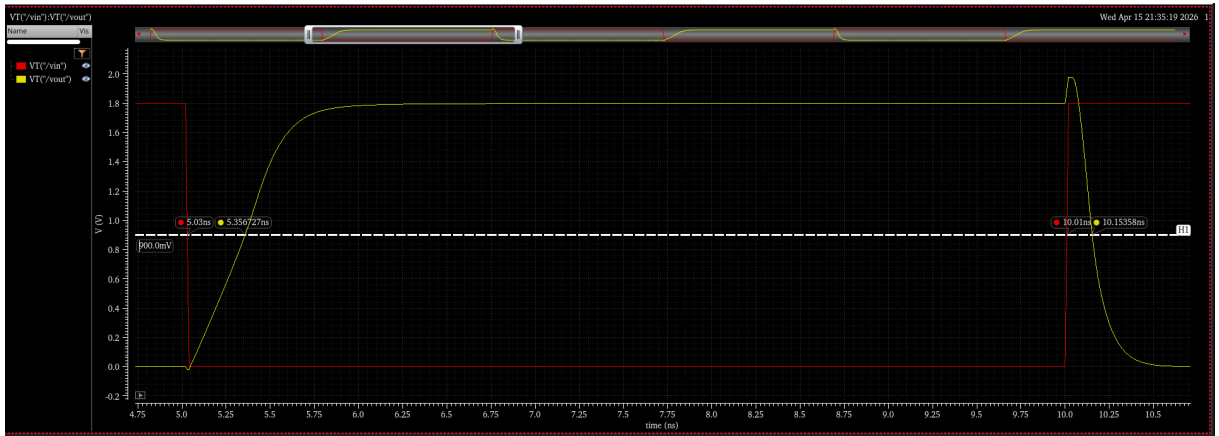


Figure 4: NAND4 Schematic Simulation. Conditions: Input transition time = 20 ps, Fanout capacitance = 10 fF.



Figure 5: NAND4 Schematic Simulation. Conditions: Input transition time = 20 ps, Fanout capacitance = 50 fF.

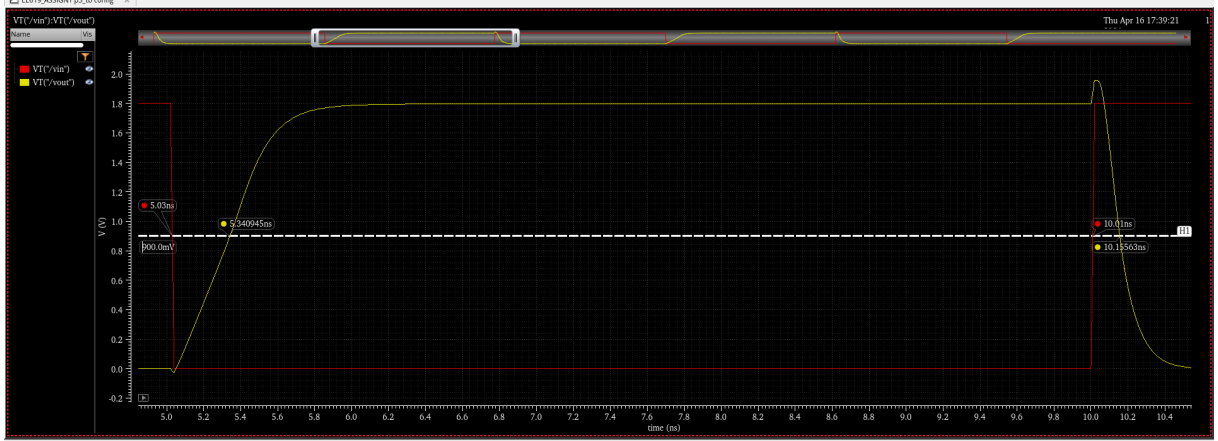


Figure 6: NAND4 Post-Layout (PEX) Simulation. Conditions: Input transition time = 20 ps, Fanout capacitance = 10 fF.

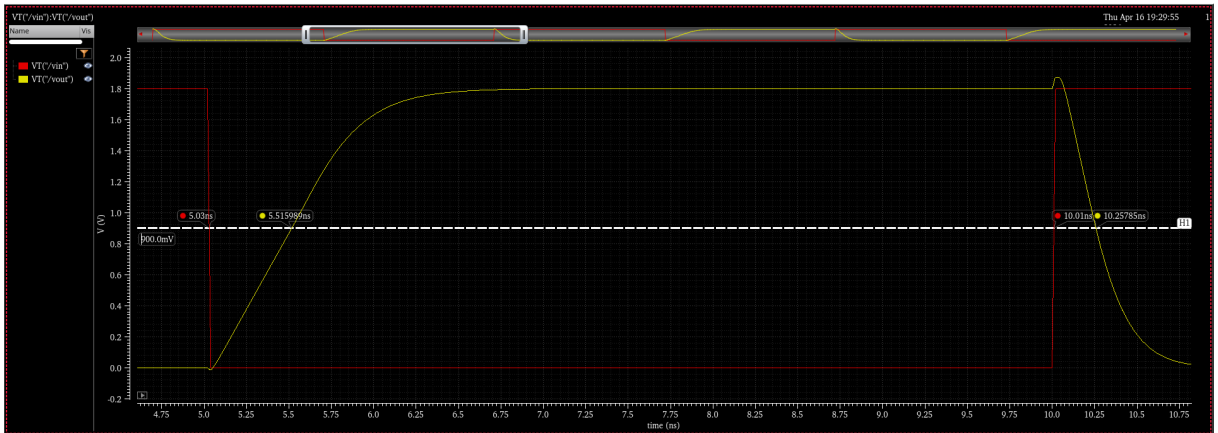


Figure 7: NAND4 Post-Layout (PEX) Simulation. Conditions: Input transition time = 20 ps, Fanout capacitance = 50 fF.

4 Second Standard Cell: NOR4

Designer: Arava Prema Chandra Reddy

4.1 Schematic and Transistor Sizing

The NOR4 gate ($F = \overline{A + B + C + D}$) was implemented in a single stage using complementary static CMOS logic.

To match the worst-case delay of the reference inverter, the transistor sizes were scaled as follows:

- **NMOS Sizing:** The pull-down network consists of 4 NMOS transistors in parallel. In the worst-case scenario, only one NMOS turns on. Therefore, the NMOS width remains the same as the reference inverter. NMOS Width = 720 **nm**.
- **PMOS Sizing:** The pull-up network consists of 4 PMOS transistors in series. To maintain the same equivalent resistance as the reference inverter's single PMOS, the width was scaled by a factor of 4. PMOS Width = 8640 **nm**.

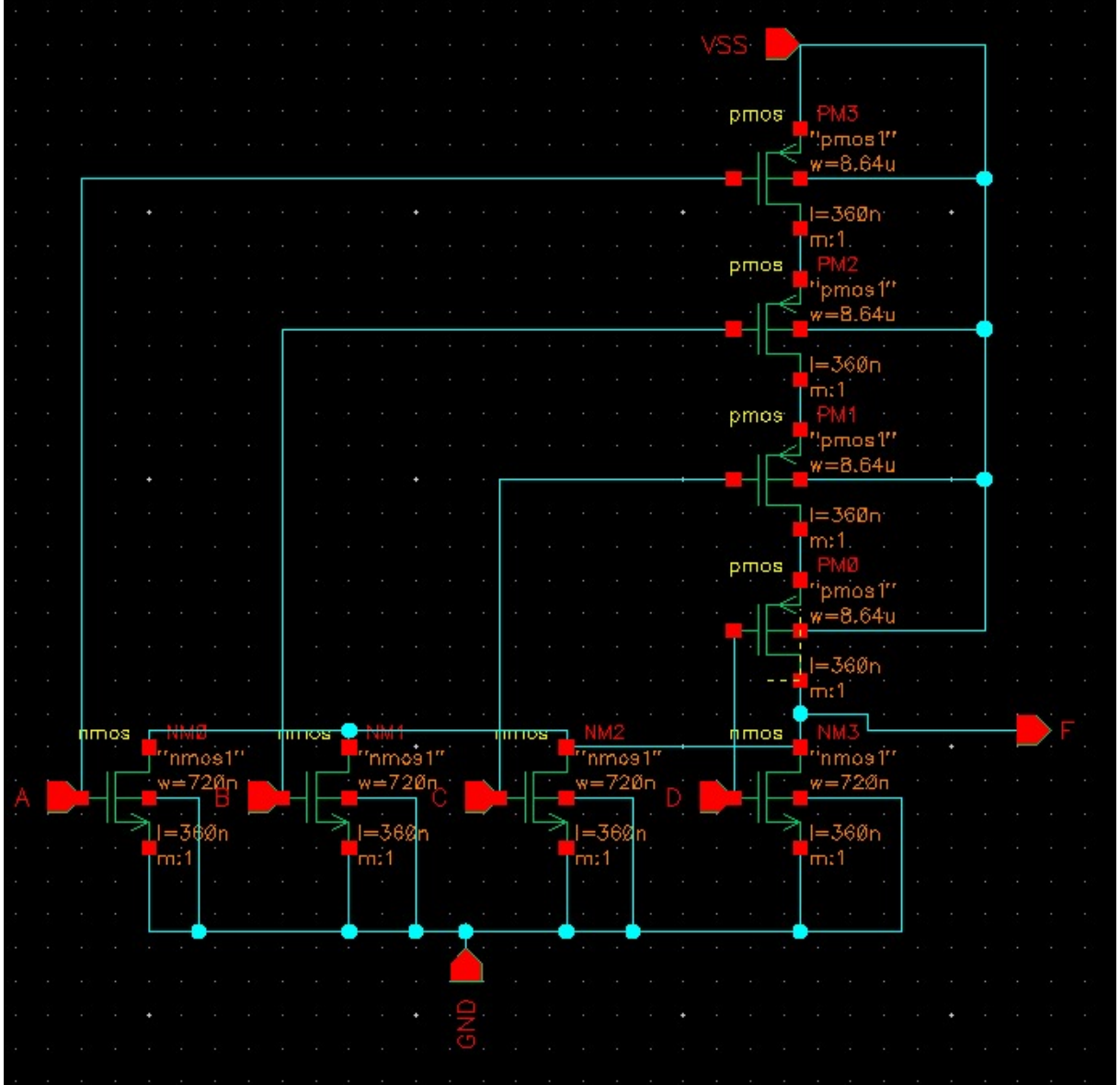


Figure 8: Schematic of the NOR4 standard cell.

4.2 Layout and Verification

The layout was designed adhering to the same cell height and VDD/GND rail widths established during the NAND4 design to ensure standard cell compatibility.

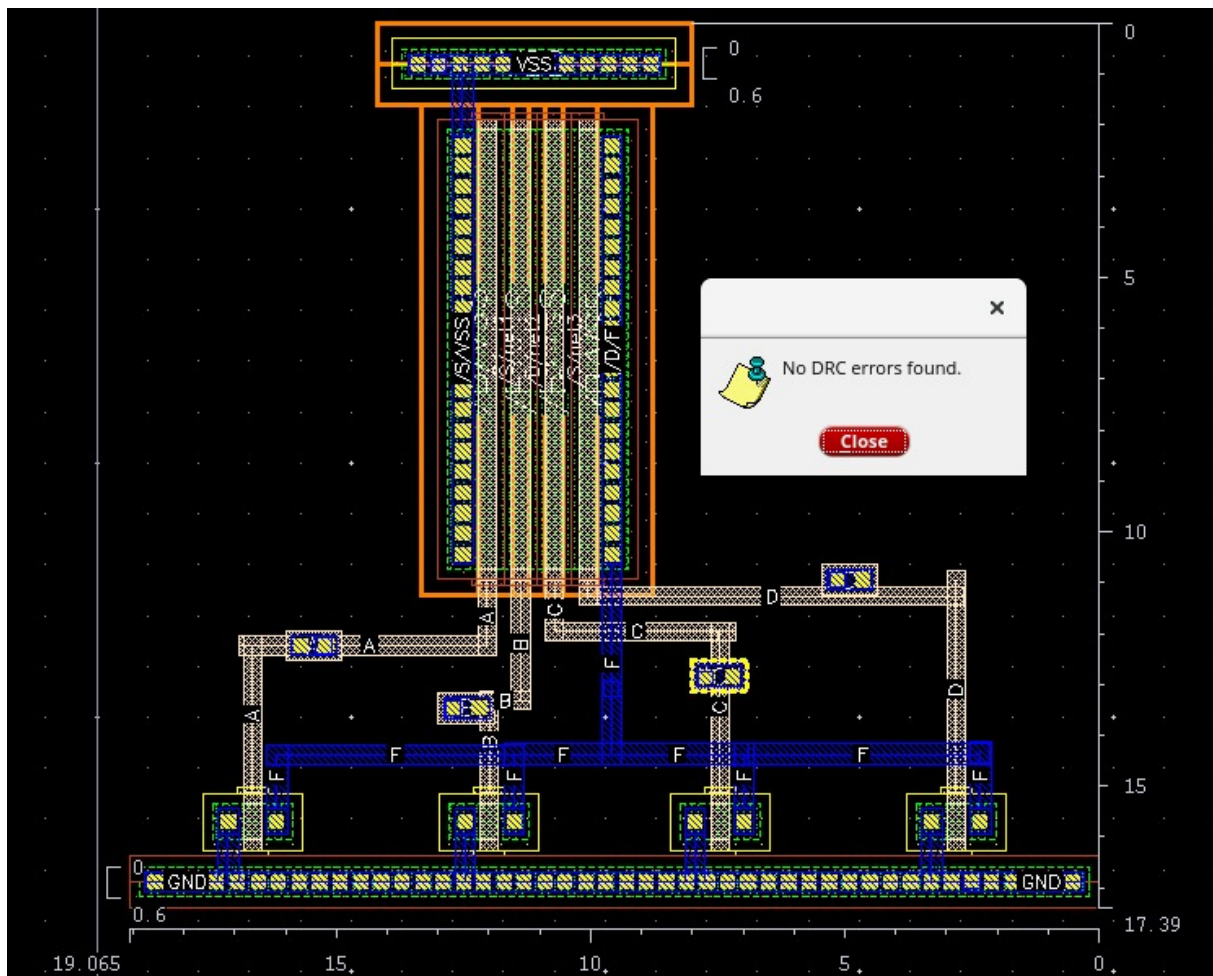


Figure 9: Layout of the NOR4 cell with clean DRC log. Height and width are explicitly marked.

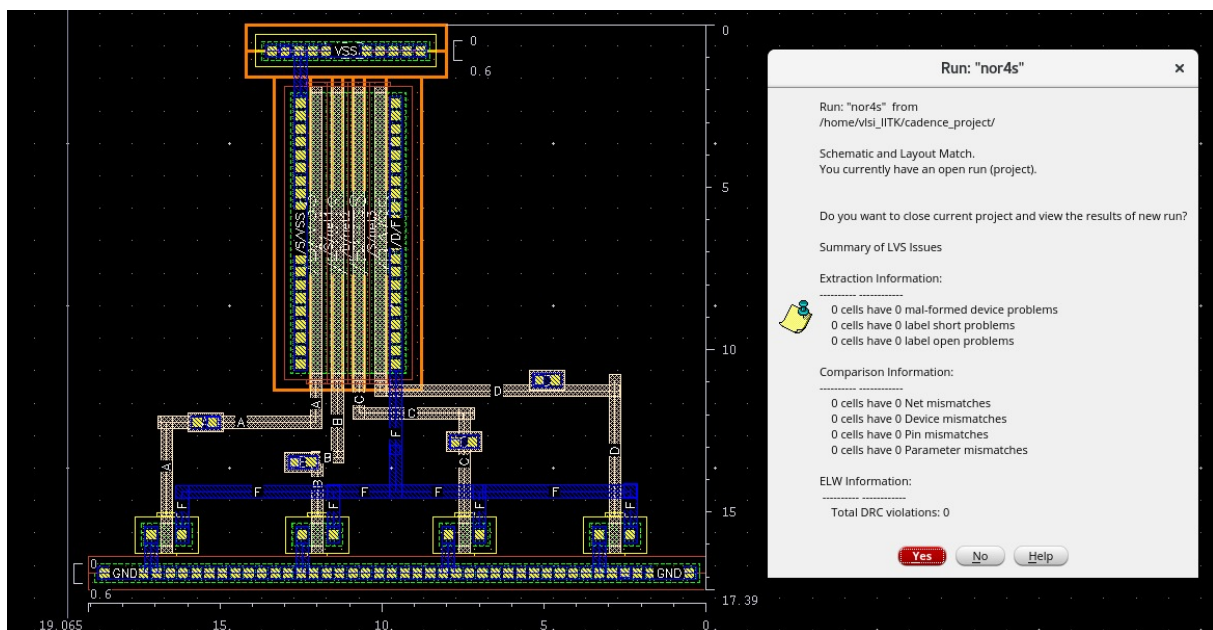


Figure 10: Layout of the NOR4 cell with clean LVS log.

4.3 Propagation Delay Analysis

4.3.1 Schematic Simulation Delays

t_{pLH} from A to F ($ABCD : 1000 \rightarrow 0000$)			t_{pHL} from A to F ($ABCD : 0000 \rightarrow 1000$)		
Load capacitance	10 fF	50 fF	Load capacitance	10 fF	50 fF
Input transition			Input transition		
20 ps	0.330 ns	0.509 ns	20 ps	0.461 ns	0.628 ns
100 ps	0.344 ns	0.525 ns	100 ps	0.465 ns	0.628 ns

Table 3: Propagation delay from A to F in NOR4 based on schematic simulations.

4.3.2 Post-Layout (PEX) Simulation Delays

t_{pLH} from A to F ($ABCD : 1000 \rightarrow 0000$)			t_{pHL} from A to F ($ABCD : 0000 \rightarrow 1000$)		
Load capacitance	10 fF	50 fF	Load capacitance	10 fF	50 fF
Input transition			Input transition		
20 ps	0.321 ns	0.496 ns	20 ps	0.430 ns	0.584 ns
100 ps	0.335 ns	0.511 ns	100 ps	0.434 ns	0.588 ns

Table 4: Propagation delay from A to F in NOR4 based on post-layout simulations.

4.4 Transient Analysis Waveforms

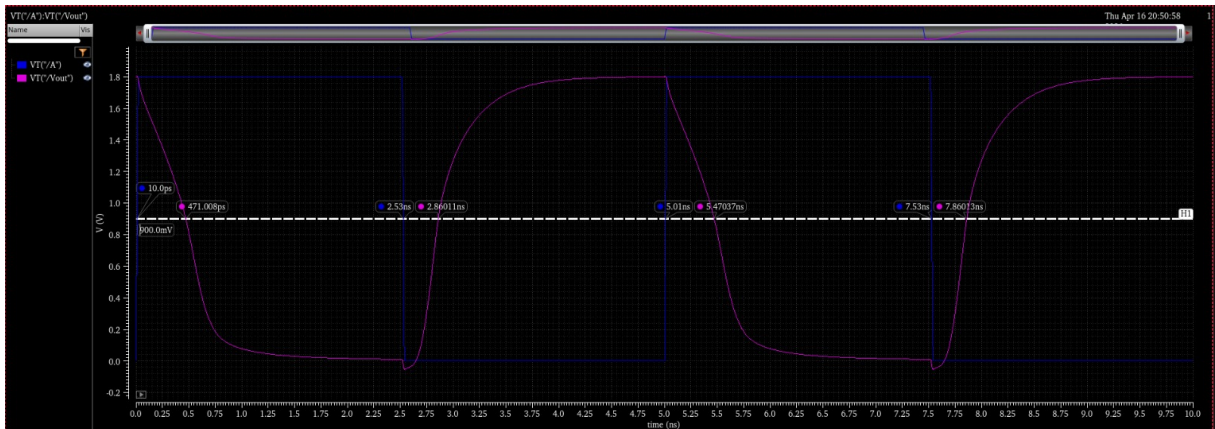


Figure 11: NOR4 Schematic Simulation. Conditions: Input transition time = 20 ps, Fanout capacitance = 10 fF.

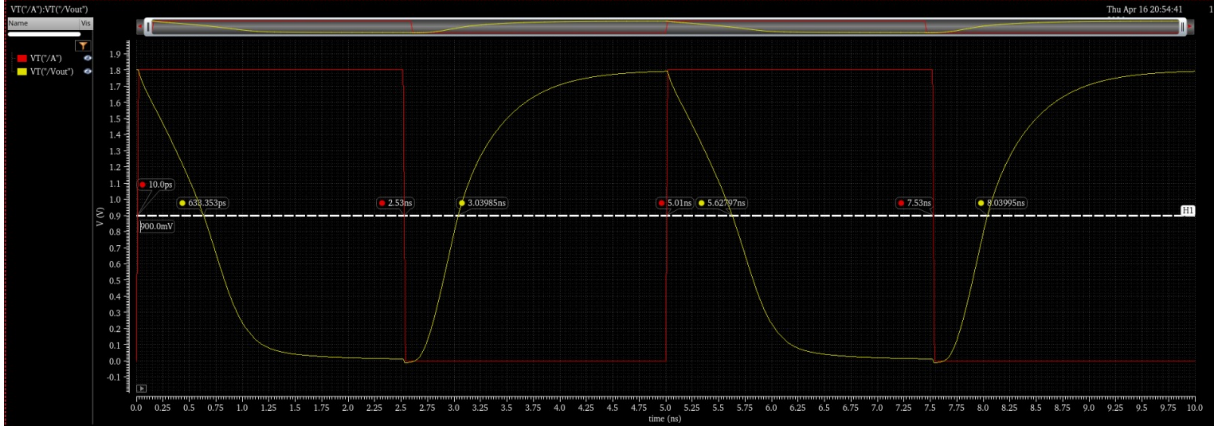


Figure 12: NOR4 Schematic Simulation. Conditions: Input transition time = 20 ps, Fanout capacitance = 50 fF.

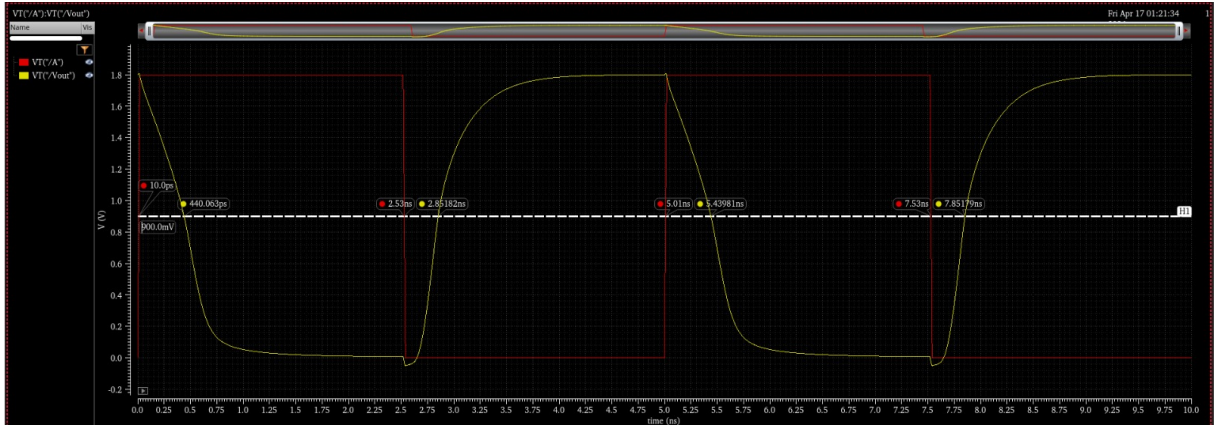


Figure 13: NOR4 Post-Layout (PEX) Simulation. Conditions: Input transition time = 20 ps, Fanout capacitance = 10 fF.

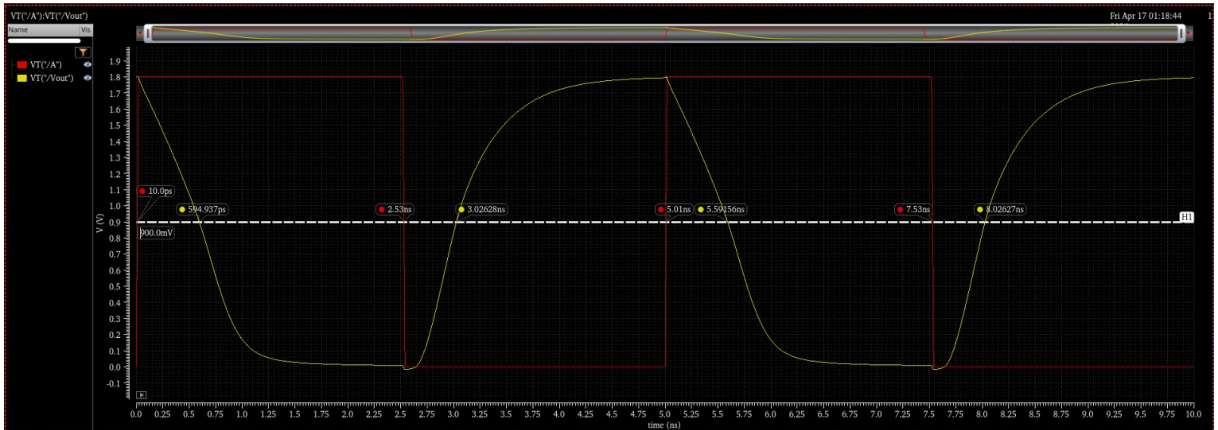


Figure 14: NOR4 Post-Layout (PEX) Simulation. Conditions: Input transition time = 20 ps, Fanout capacitance = 50 fF.